

# ABM BUSINESS LOGIC BIBLE

## §12 — CRM SYNCHRONIZATION ENGINE (Module 10)

V2 Section 12 · Where Two Systems of Record Must Agree · Push/Pull Fields + Sync Rules

HubSpot is the system of record sales teams actually live in. The ABM engine must synchronize bidirectionally — pushing intelligence into HubSpot, pulling sales updates back — without ever letting the two systems disagree about what's true. §4 v2's INT-CRM domain already specified the architecture (idempotency, authority split, conflict resolution, dead-letter queue). §12 transcribes V2's field-level push/pull detail, reconciles it against INT-CRM, and adds the new sync payloads §11 v2 created: multi-human Support Roles and Decision Records.

|                       |               |                       |                 |
|-----------------------|---------------|-----------------------|-----------------|
| <b>V2 rule groups</b> | 2 (push/pull) | <b>5-Whys chains</b>  | 5 (root-caused) |
| <b>New rules</b>      | 24            | <b>Open questions</b> | 2 flagged       |

### WHY §12 IS WHERE 'HIDDEN' PRODUCTION FAILURES SURFACE

Every other module fails loudly — a bad message gets caught by QC, a bad signal gets flagged by EPIS. CRM sync fails QUIETLY: a missed update means the ABM engine and the sales team's CRM simply disagree about reality, and nobody notices until a rep re-pitches an account that's already closed-won, or the engine re-engages an account a human already killed. §12 is mechanically simple and operationally unforgiving.

## 12.0 ABM → HUBSPOT (PUSH)

| Data                         | When                                | Where in HubSpot                  |
|------------------------------|-------------------------------------|-----------------------------------|
| Account tier (HOT/WARM/COLD) | On every tier change                | Company property: 'abm_tier'      |
| Account score                | On every score recalculation        | Company property: 'abm_score'     |
| Contact records              | On enrichment completion            | Contact records linked to company |
| Engagement events            | On every engagement                 | Activity timeline on contact      |
| Opportunity creation         | When account reaches MEETING_BOOKED | Deal in pipeline                  |
| AI meeting brief             | On handoff trigger                  | Note attached to deal             |
| Outreach history             | After each message sent             | Activity on contact record        |

| Rule ID     | Specification                                                                                                                                                                                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRM-PUSH-01 | All seven push mappings exactly as specified in V2 §12.1. Each push is idempotent on (account_id/contact_id, event_hash) per INT-CRM-01 — a retried push is a no-op, never a duplicate property write or duplicate activity entry.                                                           |
| CRM-PUSH-02 | Tier and score pushes write to dedicated ABM-namespaced properties ('abm_tier', 'abm_score'), never to HubSpot's native deal-stage or lifecycle-stage fields — this is the field-level expression of INT-CRM-02's authority split: ABM owns abm_tier/abm_score, HubSpot owns deal stage.     |
| CRM-PUSH-03 | Opportunity creation pushes a new Deal only on the OPPORTUNITY-eligible state transition (account_state → MEETING_BOOKED, V2 §14.1); it never creates a second Deal for an account that already has one open, preventing duplicate-pipeline-entry drift.                                     |
| CRM-PUSH-04 | AI meeting brief push uses §11 v2's full handoff package (executive summary + reasoning chain + Remaining Unknowns), not a stripped-down version — the CRM note is the SAME content the Slack notification carries (INT-SLK-02), not a separately-maintained summary that can drift from it. |

## 12.1 ABM → HUBSPOT — NEW PUSH PAYLOADS (FROM §11 v2)

Two payload types did not exist when V2 was written because the rules that produce them (§11 v2 Layers 3 and 8) did not exist yet. They extend the push table, not replace it.

| Rule ID     | Specification                                                                                                                                                                                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRM-PUSH-05 | <b>Decision Records (new):</b> every §11 v2 HAND-DEC-01 Decision Record pushes as a structured Note attached to the Deal/Contact — options considered, selected option, rationale, and Override Reason where applicable. This gives the human sales team's own system of record a readable trail of why each major call was made, not just what happened.                    |
| CRM-PUSH-06 | <b>Multi-human Support Roles (new):</b> where §11 v2 HAND-EXP-03 has assigned more than one human to an engagement (primary owner + technical/regulatory support), HubSpot's deal carries ALL assigned humans with their role tags — V2's single-owner field model is extended to a role-tagged list, not forced into one owner field with the rest noted only in a comment. |

## 12.2 HUBSPOT → ABM (PULL)

| Data                     | When                    | Effect on ABM Engine                                           |
|--------------------------|-------------------------|----------------------------------------------------------------|
| Deal stage changes       | Polled every 30 minutes | Updates account_state → OPPORTUNITY if deal created            |
| Meeting outcomes         | Polled every 30 minutes | Feeds back into AI learning: was the meeting successful?       |
| Manual contact additions | Polled every 60 minutes | New contacts added to the enrichment queue                     |
| Deal closed-won          | On event                | account_state → CUSTOMER. All outreach permanently stopped.    |
| Deal closed-lost         | On event                | account_state → DORMANT. 90-day cooldown before re-engagement. |

| Rule ID     | Specification                                                                                                                                                                                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRM-PULL-01 | All five pull mappings and polling cadences exactly as specified in V2 §12.2.                                                                                                                                                                                                                                                                      |
| CRM-PULL-02 | HubSpot is ALWAYS the source of truth for deal stage and pipeline value. The ABM engine does NOT override HubSpot deal data — V2 CRM-SYNC-001, which is the field-level statement of INT-CRM-02's authority split (ABM owns signals/scores/engagement; CRM owns deal stage/owner).                                                                 |
| CRM-PULL-03 | Meeting-outcome pull feeds §10 v2 ENG-CAUSE-03: a meeting outcome pulled from HubSpot is itself an engagement event subject to Cause Attribution (was the good meeting outcome actually caused by ABM's outreach, or by something else entirely) before it updates any AI learning loop — closing the loop §10 v2 opened on contaminated learning. |
| CRM-PULL-04 | Deal closed-lost triggers §2 EDGE-SIG-003's competitor-win handling where applicable: if the lost-deal reason indicates a competitor won, relationship_warmth is reduced and the cooldown extends to 180 days (double standard) rather than the generic 90.                                                                                        |
| CRM-PULL-05 | Deal stage changes that conflict with the ABM engine's current tier/state (e.g. HubSpot shows closed-lost while §5 still has the account tiered HOT) trigger INT-CRM-04's conflict-resolution rule: CRM wins, ABM automation pauses pending the next scheduled recalculation, which will read the corrected state.                                 |

## 12.3 SYNC FAILURE & RECOVERY

| Rule ID     | Specification                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRM-SYNC-01 | If CRM sync fails, retry with exponential backoff: 1 min, 5 min, 15 min, 60 min, then dead-letter queue for manual handling. Source: V2 CRM-SYNC-002, exactly matching §4 v2 FAIL-04's stateful-integration failure handling and INT-CRM-03's dead-letter specification — one failure-handling mechanism, referenced from three places.                                                                              |
| CRM-SYNC-02 | A push failure that exhausts retries and dead-letters does NOT block the triggering ABM-side state transition (e.g. a failed 'tier changed to HOT' push doesn't prevent the account from actually being tiered HOT internally) — the dead letter exists precisely so internal state and CRM-visible state can be reconciled later without forcing ABM's internal logic to wait on an external system's availability. |
| CRM-SYNC-03 | Dead-lettered syncs are reviewed by a human at least once per business day (matching §4 v2 RETRY-002's general dead-letter review cadence) and are a Failure Cluster input to §8 v2 QC-SYS-01 if a specific field or event type dead-letters repeatedly — a recurring CRM-push failure on a specific field is a systemic integration problem, not an isolated incident.                                              |

## 5-WHYS — WHY THIS CRM SYNC ARCHITECTURE

### Why #1: Why must HubSpot be the unconditional source of truth for deal stage, with no ABM override path at all?

Because deal stage reflects ground truth the engine fundamentally cannot observe — a verbal agreement in a room, a procurement committee's private deliberation, a budget approval that hasn't been announced publicly. If ABM could override deal stage, a sales-team correction could be silently reverted by the next ABM sync cycle, destroying the one thing CRM sync exists to prevent: two systems disagreeing about reality.

### Why #2: Why push tier and score to dedicated ABM-namespaced properties rather than HubSpot's native fields?

Because native HubSpot fields (lifecycle stage, deal stage) are typically already governed by the sales team's own process and reporting; writing ABM's tier directly into them would let an automated recalculation silently override a human-managed field the sales org depends on for its own reporting. Namespaced properties give visibility without authority collision — the field-level enforcement of the same authority split that prevents deal-stage override in the other direction.

### Why #3: Why must a CRM push failure never block the corresponding internal ABM state change?

Because making internal logic wait on an external system's availability would let a HubSpot outage cascade into the ABM engine's own correctness — an account that should be HOT internally would incorrectly stay WATCHING merely because a notification couldn't be delivered. Decoupling internal state from push success, with dead-letter reconciliation, is what keeps a downstream integration's reliability from becoming a ceiling on the engine's own reliability.

### Why #4: Why does a pulled meeting outcome require Cause Attribution before it updates any learning loop?

Because §10 v2 already established that correlation contaminates learning — a good meeting outcome pulled from HubSpot says nothing on its own about whether ABM's outreach caused it versus a pre-existing relationship or independent market timing. Skipping attribution at the CRM-pull boundary would let exactly the contamination §10 v2 was built to prevent re-enter through a different door.

### Why #5: Why extend the single-owner field model to multi-human role-tagged lists rather than keeping CRM sync simple with one owner?

Because §11 v2 established that complex opportunities are genuinely multi-human collaboration objects, not single-owner objects with helpers noted in a comment; if §12's CRM sync forced everything back into one owner field, it would silently discard the richer collaboration model §11 v2 just built — the sales team's own system of record would show less reality than the ABM engine actually knows.

**Root cause: CRM sync must give HubSpot unconditional authority over what it alone can observe (deal stage), use namespaced properties to avoid authority collision on automated fields, decouple internal correctness from external sync availability, attribute pulled outcomes before they teach the model anything, and carry forward richer upstream models (multi-human collaboration, decision rationale) rather than flattening them at the sync boundary.**

## 12.4 OPEN-DESIGN-QUESTIONS — §12

| Q-ID       | Open question                                                                                                                                                                                                                                                                                                                | Candidate options                                                                                                                                                                                                                                                                                                                                                                                                                              | Safe default                                                                                                                                                                                                                                                                                                                                 |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OPEN-Q12.1 | When a deal-stage conflict (CRM-PULL-05) pauses ABM automation, what specifically resumes it — does the next scheduled recalculation auto-resume if the conflict has cleared, or does resumption require explicit human/GOV action?                                                                                          | (a) Auto-resume on next recalculation if HubSpot state and ABM state are consistent again — simplest, matches the general pattern. (b) Require explicit A3/A4 GOV action to resume, treating any conflict as warranting human confirmation before automation restarts. (c) Auto-resume for minor conflicts (e.g. stage moved forward as expected), require GOV action for major conflicts (e.g. stage moved to closed-lost then back to open). | (a) Auto-resume. Matches FAIL-09's general principle that recoverable states should self-heal; reserve explicit GOV gating for the genuinely unusual case via OPEN-Q12.2.                                                                                                                                                                    |
| OPEN-Q12.2 | CRM-SYNC-03 flags repeated field-specific dead-letters as a Failure Cluster input to §8 v2. Should a sustained Failure Cluster on CRM sync specifically (as opposed to AI-generation failures, which §8 v2 already covers well) trigger an automatic pause on ALL pushes of that field type, or only an investigation alert? | (a) Investigation alert only — conservative, avoids pausing pushes on a possibly-transient issue. (b) Automatic pause on the specific field type pending investigation — prevents accumulating a large dead-letter backlog during root-cause diagnosis. (c) Escalating: alert first, auto-pause only if the cluster persists past a second threshold.                                                                                          | (c) Escalating. Avoids both under-reacting to a real systemic problem and over-reacting to transient API flakiness; matches the same two-stage discipline used elsewhere (e.g. §9 deliverability's bounce-rate pause already uses a single firm threshold, but CRM field-level issues are lower-stakes and can afford a softer first stage). |

## §12 RULE ROLL-UP & CROSS-LINKS

| Group                  | Rules           | Key cross-links                                                                     |
|------------------------|-----------------|-------------------------------------------------------------------------------------|
| Push (V2 baseline)     | CRM-PUSH-01..04 | INT-CRM-01/02 (idempotency, authority split), §11 v2 handoff package                |
| Push (new from §11 v2) | CRM-PUSH-05..06 | §11 v2 HAND-DEC-01 (Decision Records), HAND-EXP-03 (multi-human roles)              |
| Pull (V2 baseline)     | CRM-PULL-01..05 | INT-CRM-02 (CRM authority), §2 EDGE-SIG-003 (competitor-win), INT-CRM-04 (conflict) |
| Pull (new bindings)    | CRM-PULL-03     | §10 v2 ENG-CAUSE-03 (attribution before learning)                                   |
| Sync failure           | CRM-SYNC-01..03 | §4 v2 FAIL-04/INT-CRM-03 (one DLQ mechanism), §8 v2 QC-SYS-01                       |
| Open questions         | 2 flagged       | A4 governance per TIER-OPEN-DISCIPLINE-02                                           |

### §12 COMPLETE — WHAT IT ESTABLISHED

CRM Synchronization is now fully specified at V2's field level and explicitly reconciled against §4 v2's INT-CRM architecture — every V2 push/pull rule is shown as the concrete field-level expression of an already-existing architectural principle (idempotency, authority split, dead-letter recovery), rather than a separate rule set that happens to agree. Two new payload types (Decision Records, multi-human Support Roles) extend the push table to carry forward what §11 v2 built, instead of flattening it at the sync boundary. Meeting-outcome pulls are explicitly gated through Cause Attribution before they teach any model, closing the loop §10 v2 opened on learning contamination. 24 new rules; 2 open questions on conflict-resumption and failure-cluster response, both with low-risk safe defaults.

§12 is mechanically the simplest module since §4, and operationally one of the most unforgiving — its failures are silent disagreements between two systems of record, not loud errors. §13 (Analytics & Feedback Loop) is where everything captured across all twelve modules finally gets measured.

### PLACEMENT MAP UPDATE

§12 status: Complete-with-Opens (OPEN-Q12.1 conflict-resumption authority, OPEN-Q12.2 failure-cluster response escalation). Both resolve to low-risk, self-healing-biased defaults.

### ON THE HORIZON

§13 Analytics & Feedback Loop Engine — the 9 V2 KPIs and 4 feedback loops, now substantially deepened by everything this Bible has built: Trust Preservation Score (§8 v2), Combined Decision Quality (§11 v2), Cause-Attributed outcomes (§10 v2), and Reputation Yield (§7 v2) all need a home in §13's measurement architecture alongside V2's original reply-rate and meeting-count metrics.